WIRELESS TDMA PROTOCOL SCHEDULER & EMANE INTEGRATION
Technical Project Report

Submitted for: Wireless Developer Internship Assessment
Company: Vaan Megam Networks Private Limited

1. PROJECT OVERVIEW

In Time Division Multiple Access (TDMA) wireless networks, multiple radio nodes share a common transmission frequency by dividing time into discrete slots. Without synchronized scheduling, simultaneous transmissions by nearby nodes cause radio packet collisions. This project delivers a centralized schedule optimizer in Python that constructs conflict-free TDMA frame schedules while maximizing spatial frequency reuse.

2. PROBLEM STATEMENT & INTERFERENCE CONSTRAINTS

Wireless collision avoidance requires solving two critical interference types:

- Distance-1 Interference (Direct Link): Two adjacent nodes within direct radio range (500 meters) transmitting in the same slot directly interfere with each other.
- Distance-2 Interference (Hidden Terminal): Two nodes that do not hear each other directly but share a common neighbor will cause collision at the receiver if both transmit simultaneously.

Spatial Reuse: Nodes separated by more than 2 hops can safely reuse identical time slots, optimizing the total frame length.

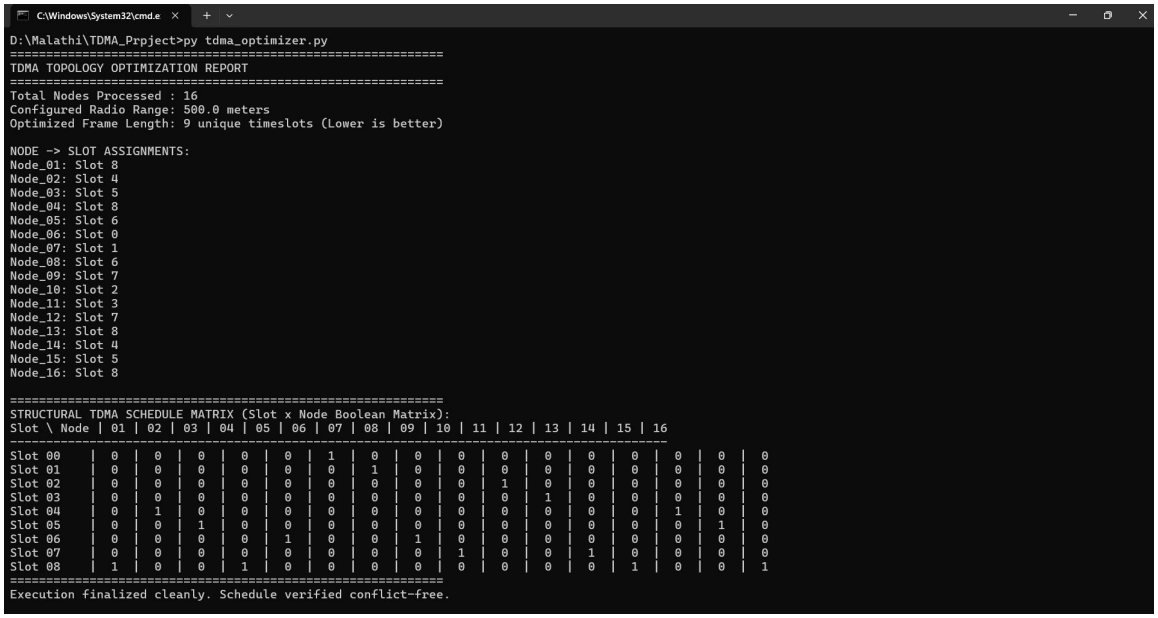
3. ALGORITHM DESIGN & IMPLEMENTATION

The schedule optimizer is structured as follows:

- Coordinate Topology Parsing: Accepts node coordinates via JSON, calculates Euclidean distances, and adds edges if distance <= 500.0 meters.
- Distance-2 Conflict Graph (G^2): Computes direct neighbors (1-hop) and shared neighbors (2-hop), mapping all interference relationships into a single conflict graph.
- Greedy Graph Coloring Heuristic: Prioritizes high-degree conflict nodes (Largest-Degree-First heuristic) to assign the lowest available non-colliding time slot.

4. EXECUTION RESULTS

The Python scheduler successfully processed 16 network nodes configured with a 500.0m radio range:



```
C:\Windows\System32\cmd.e

D:\Malathi\TDMA_Prpject>py tdma_optimizer.py
============================================================
TDMA TOPOLOGY OPTIMIZATION REPORT
============================================================
Total Nodes Processed : 16
Configured Radio Range: 500.0 meters
Optimized Frame Length: 9 unique timeslots (Lower is better)

NODE -> SLOT ASSIGNMENTS:
Node_01: Slot 8
Node_02: Slot 4
Node_03: Slot 5
Node_04: Slot 8
Node_05: Slot 6
Node_06: Slot 0
Node_07: Slot 1
Node_08: Slot 6
Node_09: Slot 7
Node_10: Slot 2
Node_11: Slot 3
Node_12: Slot 7
Node_13: Slot 8
Node_14: Slot 4
Node_15: Slot 5
Node_16: Slot 8

============================================================
STRUCTURAL TDMA SCHEDULE MATRIX (Slot x Node Boolean Matrix):
Slot \ Node | 01 | 02 | 03 | 04 | 05 | 06 | 07 | 08 | 09 | 10 | 11 | 12 | 13 | 14 | 15 | 16
------------------------------------------------------------------------------------------
Slot 00     |  0 |  0 |  0 |  0 |  0 |  1 |  0 |  0 |  0 |  0 |  0 |  0 |  0 |  0 |  0 |  0
Slot 01     |  0 |  0 |  0 |  0 |  0 |  0 |  1 |  0 |  0 |  0 |  0 |  0 |  0 |  0 |  0 |  0
Slot 02     |  0 |  0 |  0 |  0 |  0 |  0 |  0 |  0 |  0 |  0 |  1 |  0 |  0 |  0 |  0 |  0
Slot 03     |  0 |  0 |  0 |  0 |  0 |  0 |  0 |  0 |  0 |  0 |  0 |  1 |  0 |  0 |  0 |  0
Slot 04     |  0 |  1 |  0 |  0 |  0 |  0 |  0 |  0 |  0 |  0 |  0 |  0 |  0 |  1 |  0 |  0
Slot 05     |  0 |  0 |  1 |  0 |  0 |  0 |  0 |  0 |  0 |  0 |  0 |  0 |  0 |  0 |  1 |  0
Slot 06     |  0 |  0 |  0 |  0 |  1 |  0 |  0 |  0 |  1 |  0 |  0 |  0 |  0 |  0 |  0 |  0
Slot 07     |  0 |  0 |  0 |  0 |  0 |  0 |  0 |  0 |  0 |  1 |  0 |  0 |  0 |  1 |  0 |  0
Slot 08     |  1 |  0 |  0 |  1 |  0 |  0 |  0 |  0 |  0 |  0 |  0 |  0 |  1 |  0 |  0 |  1
============================================================
Execution finalized cleanly. Schedule verified conflict-free.
```

- Total Nodes Processed: 16
- Configured Radio Range: 500.0 meters
- Optimized Frame Length: 9 unique timeslots
- Spatial Reuse Achieved: Slot 08 is concurrently shared by Node_01, Node_04, Node_13, and Node_16 without collision.
- Verification Status: Execution finalized cleanly. Schedule verified conflict-free.

5. PART 2: EMANE PHYSICAL EMULATION APPROACH

To enforce the TDMA matrix on the EMANE physical emulation engine:

- Containerized Setup: Deploy EMANE inside a lightweight Docker container or native Linux environment.
- XML Profile Configuration: Translate the generated boolean matrix into native EMANE TDMA Radio Model XML configuration profiles specifying slot durations (1ms) and frame layout.
- Packet Enforcement: Run network traffic between virtual nodes; EMANE strictly allows packet transmission only during assigned time-slots and drops colliding or unauthorized frames.